Welcome to LUG!



Please check in:
https://forms.gle/4WQqtqCyrKjHPQvMA

Coming Up

- 11/6 - SAD LAN Night (3pm - 9pm)
- 11/7 - System Source SWAP Meet and Repair Workshop (9am-1pm)
  - LUG will be tabling and selling items and accepting donations.
- 11/11: NTP! - Zach A11/18: Shane Donahue Talk

Who are we?



President
Alex (blahaj)



Vice President
dani (puppy)



Treasurer
Hamza
(scoreoutofbounds)



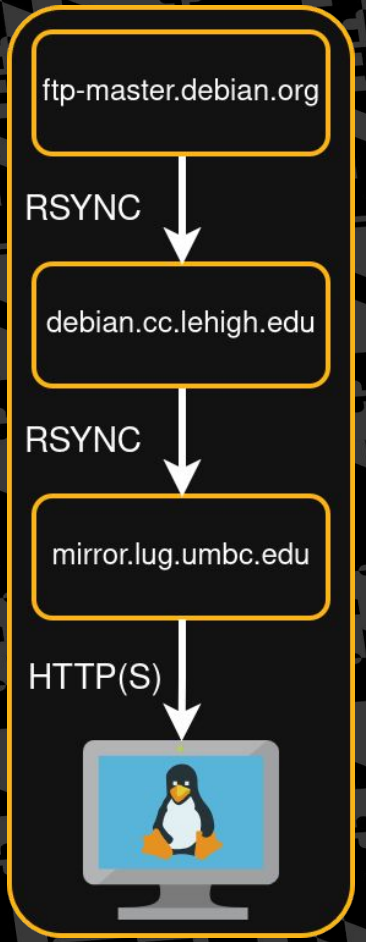
Secretary
Milk (🥛)

UMBC Linux Users Group Presents

Server Building 101

What is a Linux Mirror

- What is a Linux package distribution mirror
  - Serves data to package managers like apt, rpm, pacman, apk, etc…
  - Also good for directly downloading ISOs
  - Downstream mirrors can sync from us as well
- Data distributed with 3 protocols; HTTP, HTTPS and RSYNC
  - HTTP(S) for communication with end clients
  - RSYNC for syncing entire repositories
- Allows for bandwidth optimization.
  - Traffic otherwise destined for a more remote mirror has less distance to travel, improving download speed.



UMBC's Linux Mirror: A brief history

- Rumor has it in the late 90s and early 00s the LUG operated some sort of package mirror
- Arch mirror pitched by former President Sean in 2023
- Discussion with DoIT begins Spring 2025
- Hardware donated and scavanaged together allowing the mirror to enter beta operation in Summer 2025
- More distros get added and we start becoming officially recognized.
- Storage upgrade allowing us to mirror more distros in spring 2026

The Mirror Today

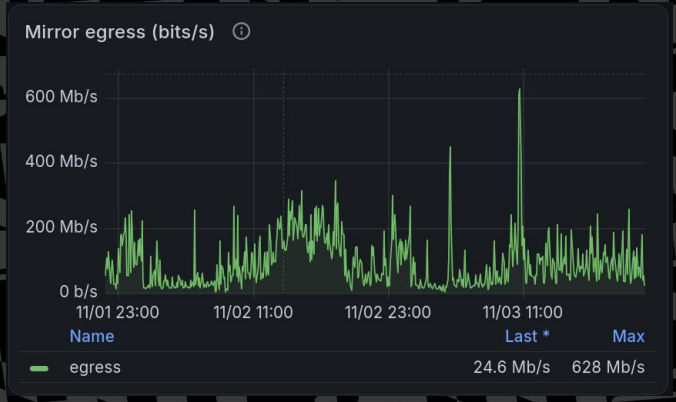
Hardware

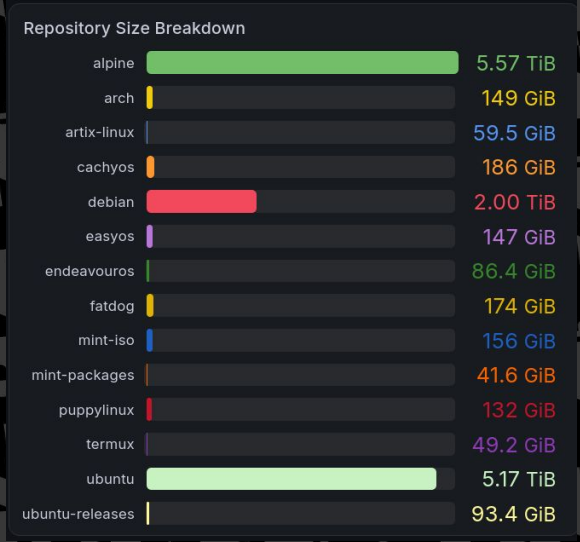
- Dell R710 (2U, Dual Socket, 160GB DDR3, ~2009)
  - Our primary hypervisor which virtualizes the services required to run the mirror operation.
- Dell Compellent SC200 (2U JBOD, ~2016)
  - JBOD (Just a bunch of Disks) functioning as bulk storage connected to the main node
- Dell Optiplex 7010
  - Runs OPNsense allowing it to be a router
  - The real fault point in our setup

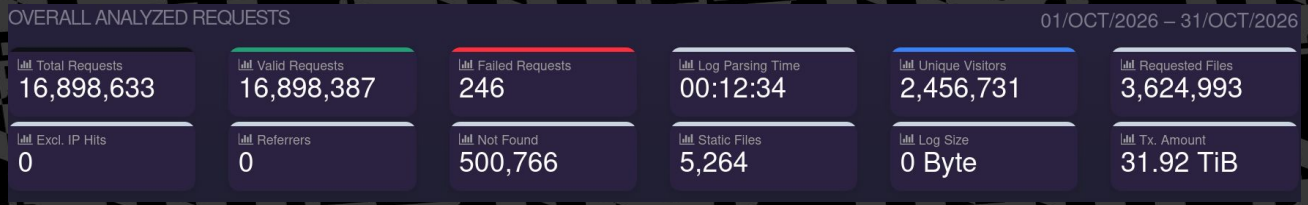


By the numbers

- Traffic >1TB/day or around a constant 100Mbps
- 14 Repositories synced totaling 14TB of data
- Highest throughput distro: CachyOS







The Mirror Tomorrow

- New server upgrade/overhaul in the works (Thanks COEIT!)
- New Server: Dell R740XD (2U, Dual Socket, 256GB DDR4)
- **Get Involved,** we are welcoming student involvement so reach out!



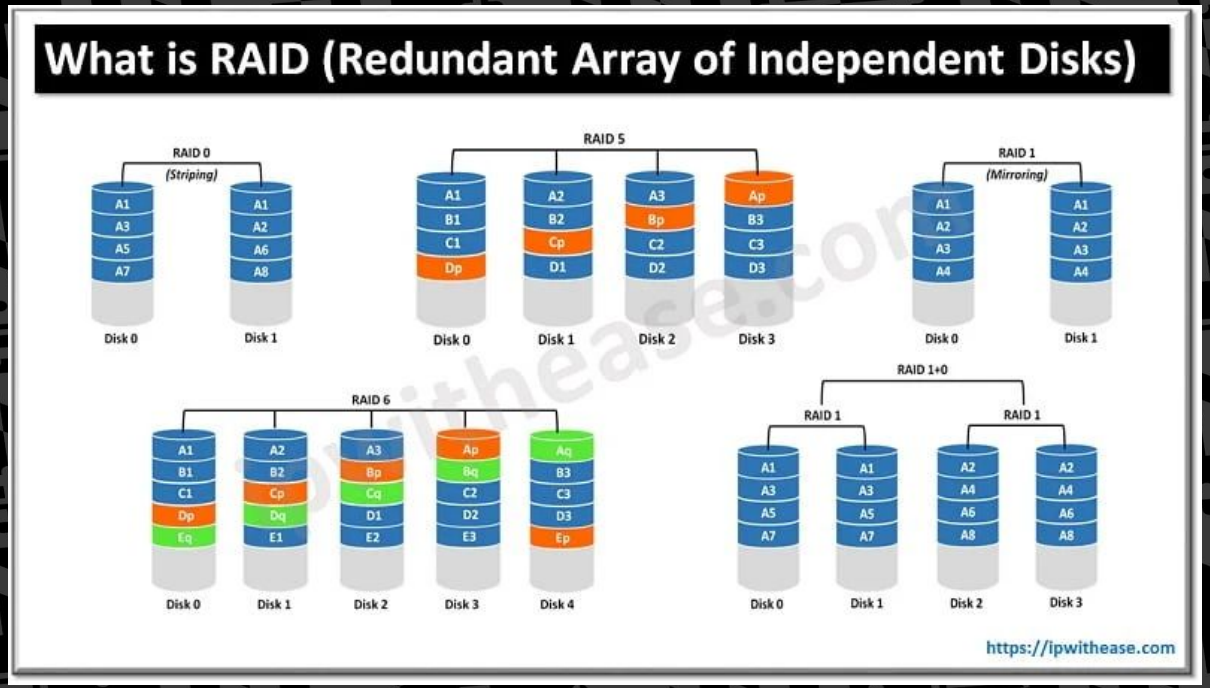


Designing for Production: Redundancy and Reliability

What makes these “enterprise” servers different from a desktop hosting a minecraft server

Redundancy and Reliability

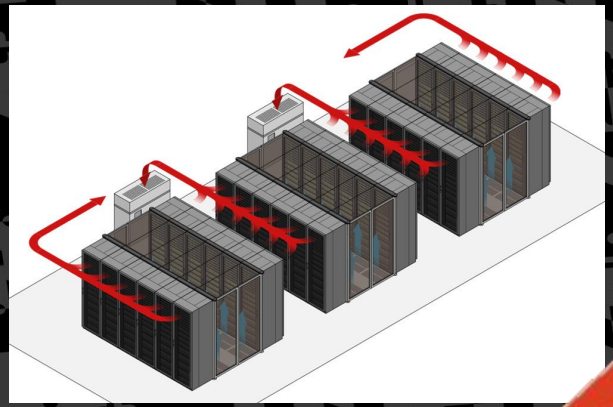
- **Data Redundancy** with RAID (Redundant Array of Independent Disks)
  - Allows for Disk failure while maintaining data integrity

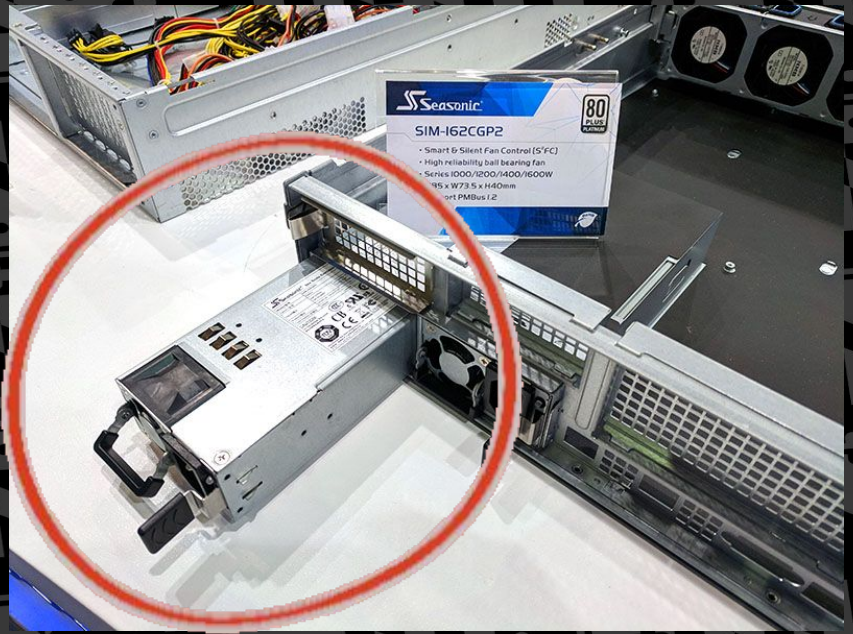


Redundancy and Reliability

Power and Environmental Reliability

- Enterprise servers have **redundant power hot-swappable power supplies.** One can fail/unplug and the server is fine.
- Thermally controlled datacenter environment, increases longevity.

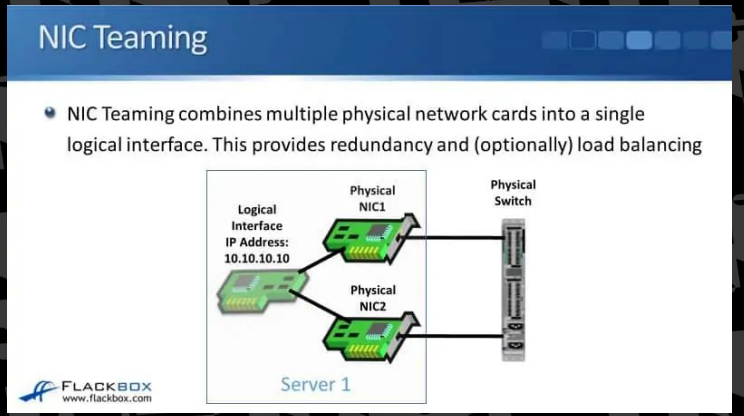




Redundancy and Reliability

Network Reliability

- High bandwidth connections prevent throttling (We have 5gbps)
  - Sometimes you may consider throttling individual clients to prevent one from overloading the server for others
- Network Load Balancing and Failover
  - Have redundant network links



Remote Management Technologies

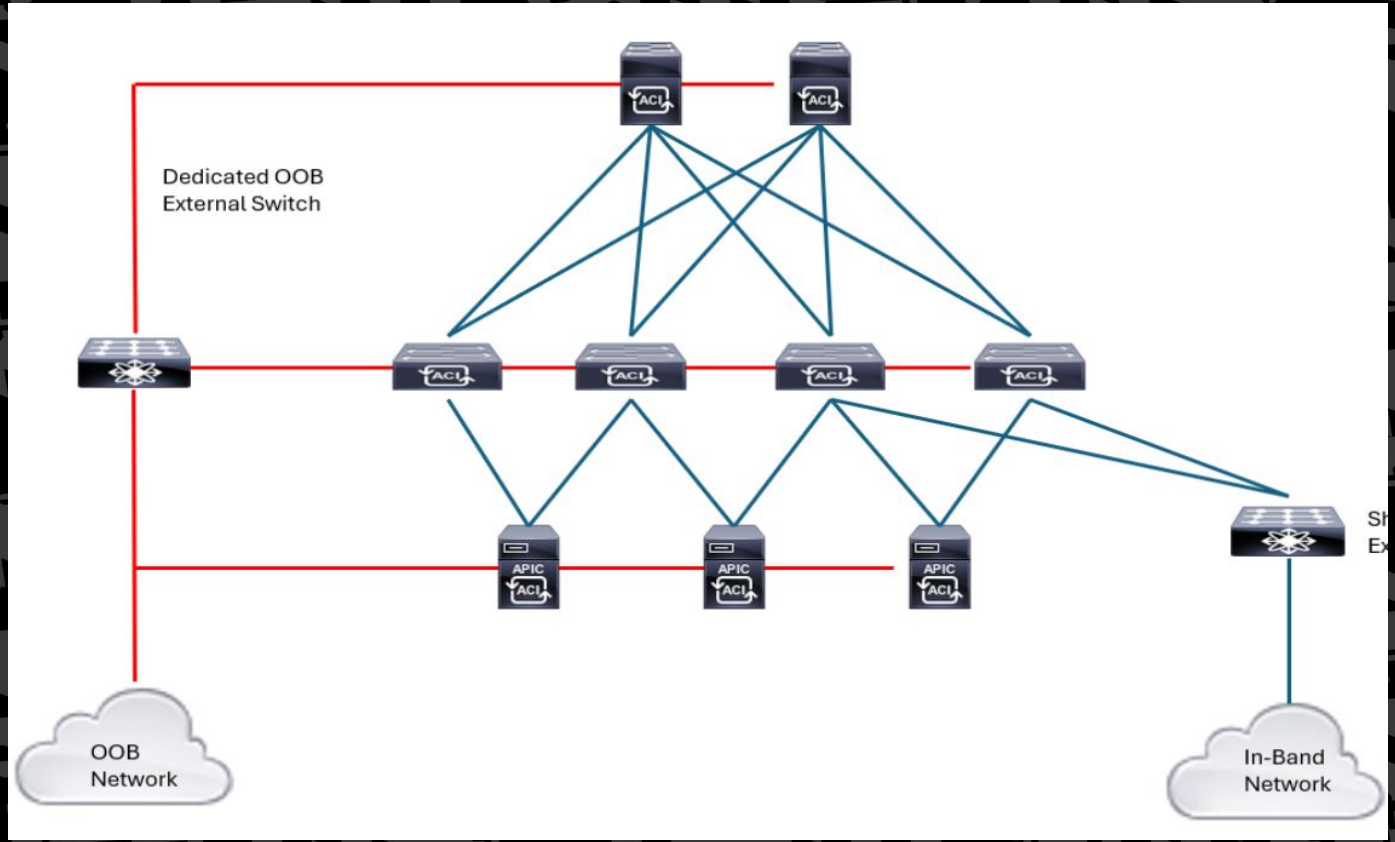
Out-of-Band-Management (OOB) or Lights-out-management (LOM) is any way to manage infrastructure remotely separate from the production network.

OOB vs. in-band: SSH or VNC are examples of in-band management.

A complete remote management system has:

- Hardware monitoring (fan speed, power voltages, etc)
- Remote power cycling
- Remote KVM (keyboard, video, mouse) and/or serial
- Local media access for disk images
- Adjust BIOS settings

Out of Band Network



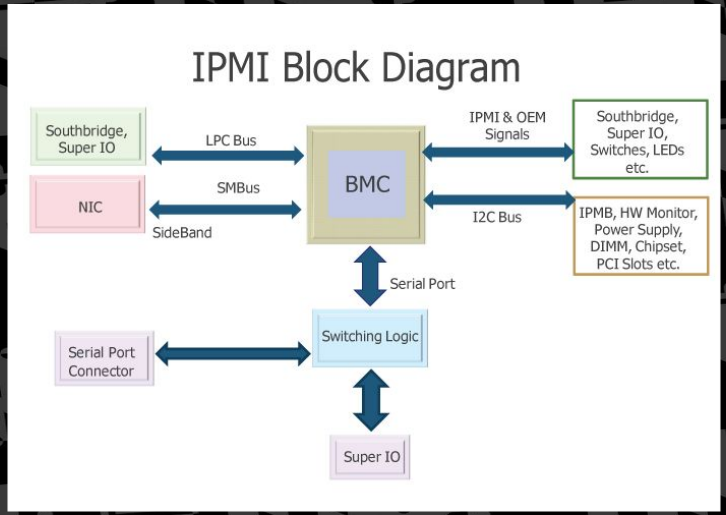
IPMI

Intelligent Platform Management Interface (IPMI) - set of specifications for an out of band management system

Controlled by the **Baseboard management controller (BMC)**



BMC Chip



Proprietary IPMI Implementations

Most manufactures develop their own version of IPMI

- Dell (i)DRAC (Integrated Dell Remote Access Controller)
- HPE iLO (Integrated Lights Out)
- Supermicro Intelligent Management

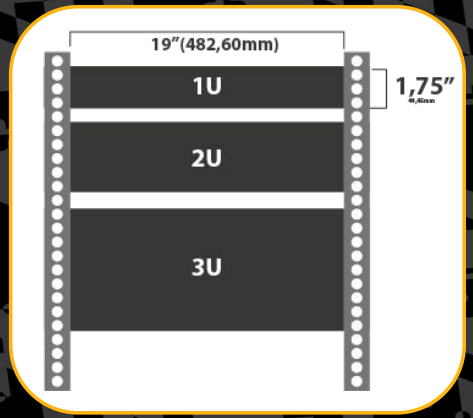
Built on top of IPMI so they still support the base standards.

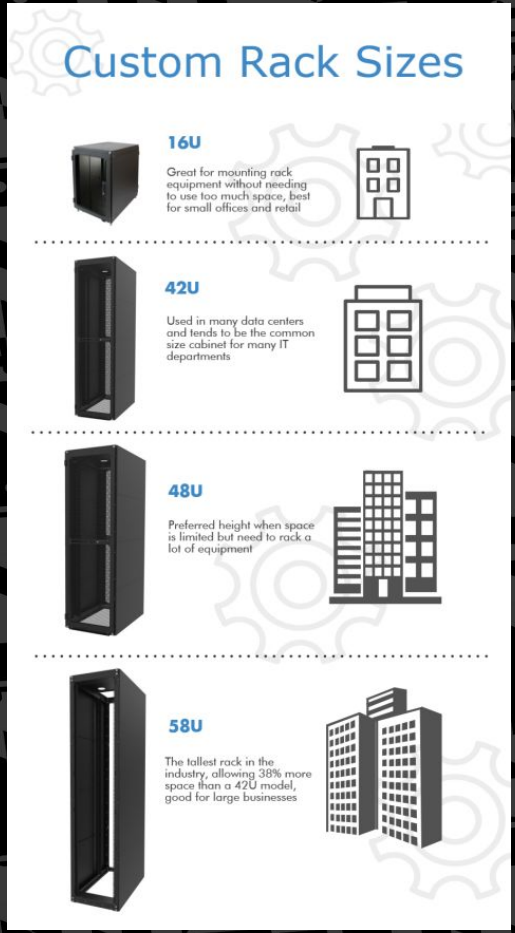
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Rack Units and Form Factor

- Instead of standard desktop chassis, enterprise equipment uses a rack-mountable standard.
- Systems measured in Rack Units (1U, 2U, 4U, etc)
- Various sizes of server racks





Consumer vs. Enterprise

	Consumer Desktop	Enterprise Server
RAM:	Standard non-ECC memory	ECC (Error-Correcting Code) RAM
Power:	Single ATX power supply	Dual hot-swappable PSUs
Management:	Local HDMI monitor & keyboard	Out-of-band IPMI / Dell iDRAC
Storage:	SATA/NVME HDDs/SSDs	SAS Drives & high-TBW enterprise SSDs/HDDs
Form Factor:	ATX tower chassis	1U/2U rackmount chassis

Operations and Maintenance

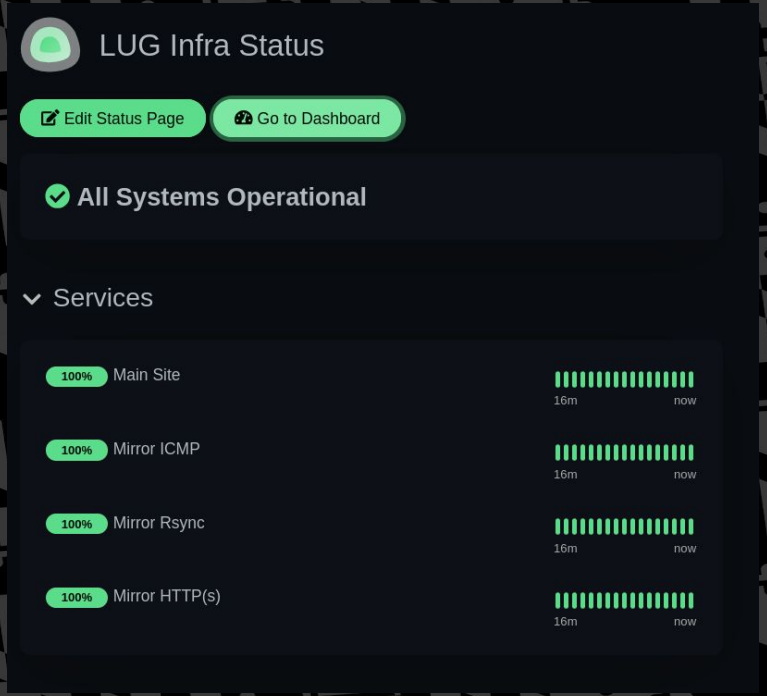
Software

- Keep your software up to date and monitor for security fixes.
  - Plan for software updates that require service downtime.
- Choose Long Term Support (LTS) releases, meaning less updates

Monitoring

Learn about an incident before it happens.

- Service status/health monitoring
  - Uptime Kuma, Nagios
- Metrics tracking
  - Prometheus + Grafana
- Log aggregation / (SIEM)
  - Watch for security events or log anomalies.
  - Splunk, Wazuh



Join us!

LUG WEBSITE



LUG DISCORD

